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# Data-driven models for electricity theft and anomalous power consumption detection: a systematic review

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**Abstract**
To maintain the effectiveness, dependability, and security of modern energy systems, analyzing and detecting anomalies in energy usage, such as electricity theft and unusual power consumption, is crucial as Smart Grid (SG) technologies become increasingly common. This survey paper comprehensively reviews the literature on energy consumption analysis and detection, focusing on detecting electricity theft and anomalous power consumption. The works that are considered in this paper are classified based on Machine Learning (ML), Deep Learning (DL), and hybrid models, to identify electricity theft and unusual power usage. Privacy preservation-based methodologies in the context of energy consumption research and summarize the survey articles. Furthermore, datasets used in electricity theft and anomalous power consumption detection, applications, challenges, and limitations related to detecting abnormal power usage and electricity theft are also discussed, and suggested future research paths to push the boundaries of this field of work. This survey study offers a thorough overview of current research trends and directions in energy consumption analysis and detection by synthesizing ideas from various studies. It benefits researchers, practitioners, and policymakers in the energy sector.

**Keywords** Anomalous power consumption · Deep learning · Electricity theft · Energy conservation · Privacy preservation

## 1 Introduction

Electricity serves as the cornerstone of modern society, powering essential services, industries, and daily activities. Its constant availability and reliability are vital for fostering socioeconomic development, driving technological innovation, and enhancing our quality of life. As global economies expand, populations grow, and technology advances, energy demands continue to rise. Consequently, identifying reliable and efficient sources of electrical energy has become increasingly critical for both present prosperity and long-term sustainability. In recent years, energy-efficient systems have gained prominence due to the urgent need to optimize energy resources and mitigate environmental impacts. Growing concerns about climate change and the finite nature of conventional energy sources have further underscored the importance of developing and adopting technologies that reduce energy consumption while promoting sustainable practices.

The exponential growth of the human population has posed complex challenges for the infrastructure and services of smart cities in meeting the needs of their inhabitants. Addressing issues such as traffic congestion, waste management, rising energy consumption, and resource overuse has become increasingly difficult. Technological advancements have not only made life more convenient but have also introduced the concept of the 'smart city,' where everything is interconnected. This concept encompasses smart services, smart transportation, smart healthcare, smart surveillance, smart buildings, and smart water and power distribution. Despite these advancements, utilities face numerous risks and challenges, one of the most significant being energy theft. Energy theft refers to the illegal use of electricity without proper payment, achieved through

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methods such as meter manipulation, feeder bypassing, and incorrect billing practices. While technology has improved metering infrastructure and introduced SG, it has also facilitated energy theft through cyber attacks enabled by information and communication technologies [1].

SGs, focused on the intelligent management of electricity generation, transmission, distribution, and control, are critical components of smart cities. An SG monitors user activities, optimizes the use of electrical resources, and ensures a long-term, sustainable power system. These features enhance efficiency, reduce energy losses, and improve security to guarantee a reliable electricity supply. Additionally, SGs contribute to environmental protection by reducing carbon emissions. SGs and smart cities collaborate to manage complex infrastructures through the integration of information and communication technology, smart services, and advanced electrical engineering. Their interaction is built on shared principles, such as integration, intelligent interconnectivity, and end-user SG components. While SGs simplify operations for their stakeholders, they also introduce new challenges. One significant issue is the threat to security and privacy, exemplified by incidents in Singapore that resulted in electricity theft. If malicious individuals gain access to private information and misuse it, the lives of consumers may be adversely affected. To ensure the smooth operation of SGs, robust security mechanisms must be implemented. Otherwise, vulnerabilities could be exploited, leading to disruptions such as power shortages, failures of heating systems, interruptions in online payment systems, and other daily inconveniences.

According to a study by the Energy Information Administration, the residential load accounted for 38.5% (1.46 trillion kWh) of the country's total annual energy consumption in 2018 [2]. Hence, effective detection of electrical load irregularities, such as electricity theft, and other types of technical or nontechnical losses in residential areas, is crucial for energy conservation and ensuring the security of a power supply system. Buildings, including heating and cooling systems, lighting, and other appliances, consume 60% of global electricity [3]. Anomalous consumption of electricity is a serious problem that requires attention. According to a report [4], non-technical losses are to be accountable for a global revenue loss of $96 billion. Studies show that electricity theft which is also a kind of anomalous power consumption costs India approximately 25% of its electricity supply, Brazil about 16%, and China roughly 6%. Electricity theft costs US utilities nearly $6 billion every year. The FBI estimated a loss of electricity theft of about $400 million per year in 2019 [5, 6].

According to estimates, the market for smart energy monitoring will be valued at $7.7 billion in 2019 and $26.9 billion in 2025. In order to encourage energy conservation and effectively manage the demand response exchange between the utility and end users, appliance-level power consumption data collection is of utmost importance. By giving end users feedback on their electricity usage, namely how they consume their electricity and the electricity consumption of their domestic equipment throughout the day, it is possible, for instance, to quickly reduce power consumption in residential households by more than 10%. Therefore, creating reliable solutions to provide end users with this knowledge and inspire them to support more sustainable energy behaviors is becoming a popular research area. The simple method to accomplish that goal is to implement "Intrusive Load Monitoring,"which entails the installation of unique smart meters/sensors at every appliance in a building, but at a significant expense. A "Non-Intrusive Load Monitoring"(NILM), which can separate the energy consumption footprints of numerous appliances using just one recording point, is another possible approach [7].

Utility companies have major problems due to electricity theft and abnormal power usage, resulting in considerable financial losses and impeding dependable power delivery. These issues affect utility companies globally. The capacity to identify and reduce electricity theft and anomalous usage patterns is essential for preserving grid stability, guaranteeing equitable billing, and improving overall energy efficiency due to modern power systems' growing complexity and size.

Need for electricity theft detection 1. **Revenue Loss Prevention:** Electricity theft leads to significant revenue losses for utility companies. Accurate detection helps in identifying unauthorized electrical power consumption, ensuring consumers are billed appropriately and revenue is not compromised.
2. **Resource Allocation:** Electricity theft can strain the power grid, leading to inefficiencies and increased maintenance costs. Detecting theft allows utilities to allocate resources more effectively and maintain the grid's reliability.
3. **Fair Distribution:** Electricity theft often places an unfair burden on law-abiding consumers who end up subsidizing the costs incurred due to theft. Detection promotes fairness in billing and distribution of resources.
4. **Regulatory Compliance:** Utility companies are required to adhere to regulations and standards. Detecting and addressing theft helps utilities maintain compliance and avoid legal and regulatory consequences.

Need for anomalous power consumption detection 1. **Energy Efficiency:** Identifying abnormal electrical

power consumption patterns helps consumers to reduce waste of energy usage.
2. **Early Problem Identification:** Anomalies in energy/electrical power consumption could signify technical problems or malfunctioning equipment. Early detection can prevent costly breakdowns and improve the lifespan of the equipment.
3. **Resource Management:** Industries can better manage energy resources by detecting irregular energy consumption patterns. This leads to cost savings, reduced environmental impact, and improved sustainability.
4. **Environmental Impact:** Abnormal energy consumption might indicate excessive usage, contributing to carbon emissions. Detection and subsequent actions can contribute to environmental conservation efforts.
5. **Demand Response:** Utilities can optimize power distribution by detecting spikes in energy consumption. This enables better peak demand management and reduces the need for additional capacity.

Electricity theft and anomalous power consumption detection ensure a reliable, efficient, sustainable energy ecosystem. These practices safeguard utility revenues and consumer equity and contribute to better resource management and environmental responsibility. Traditional models like statistic-based, rule-based, and game theory-based models can be used to detect electricity theft and anomalous power consumption. However, they have their own disadvantages as compared to data-driven models like ML and DL.

- Traditional methods rely on rules or preset models, which might be difficult to adjust to new trends or modifications in the data. Traditional models are more static than data-driven models.
- Datasets with large sizes can be challenging when using traditional methods since they can become inefficient to scale adaptively as data volumes increase. Sometimes, they need extensive manual feature selection and preprocessing.
- Many traditional models assume the linear relationship between the features and fail to recognize non-linear and complex relationships. Traditional methods are more sensitive to noise in the data, leading to wrong output.
- Traditional models may need more human intervention to supervise and modify than data-driven models. This raises the possibility of human mistakes and adds time to the process.
- As these traditional models need manual or offline processing, they are unsuitable for real-time data processing and applications.
- Effective applications involving traditional models need domain experts to set the parameters that limit their generalizability.

Due to the disadvantages mentioned above with traditional models, data-driven models that offer high accuracy, scalability, adaptability, efficiency, etc., got more attention from researchers.

In this review paper, the following research questions are addressed:

- How can data-driven approaches like ML and DL be employed to detect electricity theft and anomalous power consumption detection?
- What are the advantages and limitations of data-driven models?
- How can privacy-preserving techniques be integrated into detecting electricity theft and anomalous power consumption to ensure data security and compliance?
- What are the gaps, challenges, and limitations in electricity theft and anomalous power consumption detection?

The following are the contributions of this review paper:

- A systematic and comprehensive review has been done on recent advancements in electricity theft and anomalous power consumption detection using data-driven approaches.
- A precise taxonomy is presented to understand various data-driven models and their advantages and disadvantages.
- Various datasets used in electricity theft and anomalous power consumption detection were summarized.
- Have identified various gaps, challenges, and limitations in the existing works with respect to electricity theft and anomalous power consumption detection.
- This work provides valuable foundation for future research directions in the field of energy consumption analysis and detection of anomalous power consumption.
- To the best of our knowledge, this paper is the first to combine the detection of abnormal power usage and electricity theft, providing a thorough method for protecting and enhancing power grid operations.

This article is organized as follows: Section 2 discusses the research paper selection process and taxonomy that was followed to summarize prior work in electricity theft and anomalous power consumption detection. It also summarizes various works proposed by the different authors. Section 3 discusses different datasets used in electricity theft and anomalous power consumption detection. Nature of the anomalies are discussed in Section 4. Section 5 discusses

the application of electricity theft and anomalous power consumption detection. Gaps in the existing literature is discussed in Section 6. Challenges and limitations are discussed in Section 7. Section 8 directs future research in electricity theft and anomalous power consumption detection. Section 9 gives the concluding remarks.

## 2 Literature survey

### 2.1 Existing literature reviews

In [8], authors have surveyed electricity theft detection methods in smart meters and they are classified as ML-based and measurement mismatch-based methods. Adverse effects and political and socioeconomic factors of electricity theft were summarized. Advanced attack detection methods and privacy-preserving methods were also discussed. In [1], authors have surveyed various energy theft detection methods and also discussed the state-of-the-art technique's strengths and weaknesses. The challenges in the detection methods are highlighted that provides future research direction.

Electricity theft detection methods and data-driven attack strategies were reviewed in [9]. The supply, demand, and control chain for the grid, emphasizing energy theft and related security vulnerabilities already present in the ecosystem for SGs were discussed. In [10], authors have reviewed articles on Artificial Intelligence (AI)-based anomaly detection in building energy consumption. In particular, a thorough survey was provided, wherein a thorough taxonomy was presented to categorize current algorithms according to various modules and parameters used, including ML algorithms, feature extraction techniques, degrees of anomaly detection, computing platforms, and application scenarios.

In [11], authors have reviewed a time series data-based outlier/anomaly detection method. Their review aimed to present a thorough and organized state-of-the-art for unsupervised outlier identification methods in time series contexts. A multi-perspective review on smart anomaly identification in sensor systems was performed in [12]. The approaches were categorized that ranged from data-driven approaches (supervised learning, reinforcement learning, DL, etc.) to traditional approaches (statistical methods, time-series analysis, signal processing, etc.). Authors of [7] suggested a multi-perspective classification of current smart NILM approaches, where a thorough analysis of recent trends in the field was offered. The contributions of DL, feature extraction, computing platforms, and application scenarios for NILM development were described.

### 2.2 Current systematic literature review

A systematic literature review is conducted on recent advancements in electricity theft detection and anomalous power consumption detection. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [13] guidelines are followed to carry out the systematic literature survey. The process involves multiple stages, starting from accessing research papers in standard databases to selecting suitable articles. Databases such as ACM, Elsevier, IEEE Xplore, Springer, Taylor & Francis, and Wiley are utilized to identify relevant studies on detecting abnormal power usage and electricity theft. The search query"*Electricity Theft OR Anomalous Power Consumption AND Detection OR Identification*"is used to retrieve relevant articles. This systematic investigation focuses on studies employing machine learning (ML), deep learning (DL), and hybrid techniques. Figure 1 illustrates the PRISMA model applied in the current study. Over 100 papers on abnormal power usage and electricity theft detection, published between 2017 and 2023, are selected for systematic analysis. Articles are shortlisted based on criteria such as language, scope, full-text availability, and article type.

Based on the articles selected from the PRISMA guidelines, taxonomy on energy consumption analysis and anomaly detection is prepared. The energy consumption analysis and detection is divided into two main categories: electricity theft detection and anomalous power consumption detection. All the articles considered in this survey related to identifying electricity theft and anomalous power consumption are data-driven. These data-driven approaches are categorized into sub-categories based on the techniques used: ML, DL, and Hybrid, its taxonomy is shown in Fig. 2.

### 2.3 Statistical-based methods in electricity theft and power consumption monitoring

Statistical-based approaches play a crucial role in electricity theft detection and power usage monitoring. Techniques such as those based on Markov models are particularly effective in identifying deviations from expected behavior and capturing probabilistic relationships within data. These methods are valued for their interpretability and computational efficiency. However, their performance in complex and dynamic scenarios is often limited by their reliance on predefined assumptions about data distribution, which may not hold in real-world situations.

The authors of [14] have presented a statistical approach for detecting electricity theft in smart grids. Higher-order statistical information about power consumption was utilized to identify cyberattacks. The approach was tested on both simulated and real power consumption datasets. A

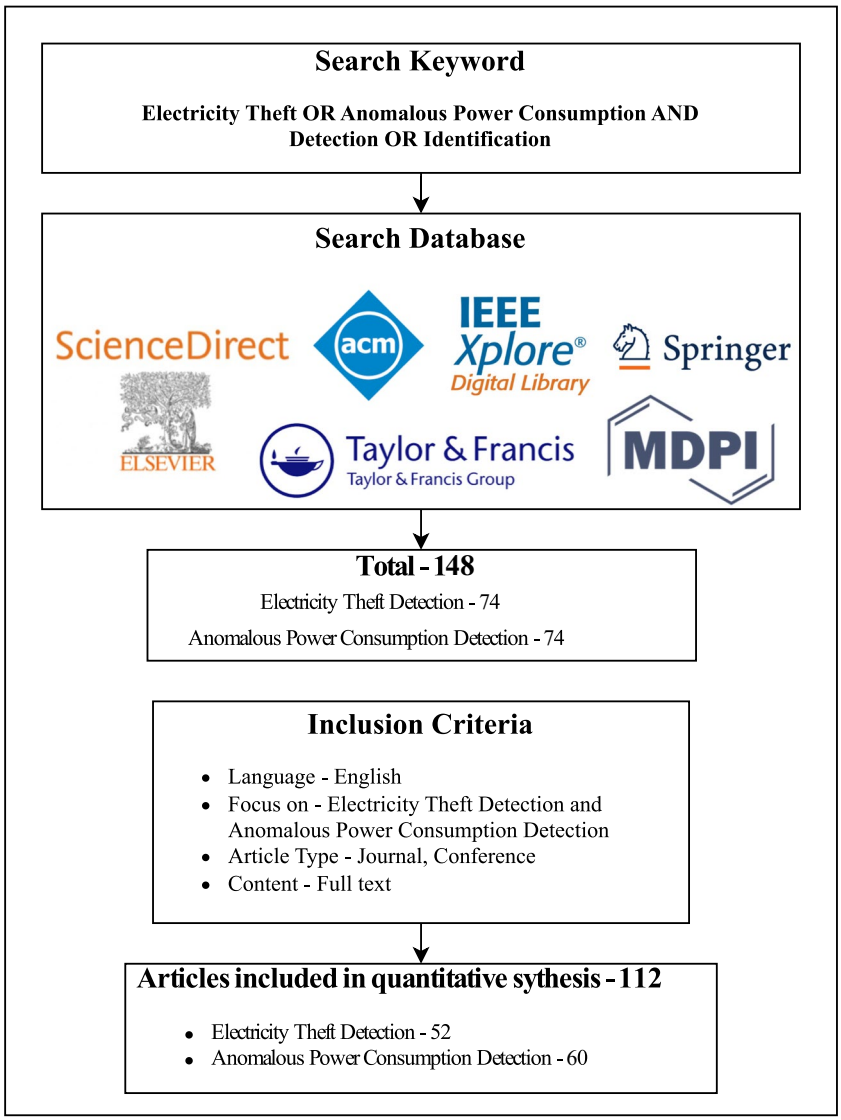



**Fig. 1** PRISMA model used for article selection

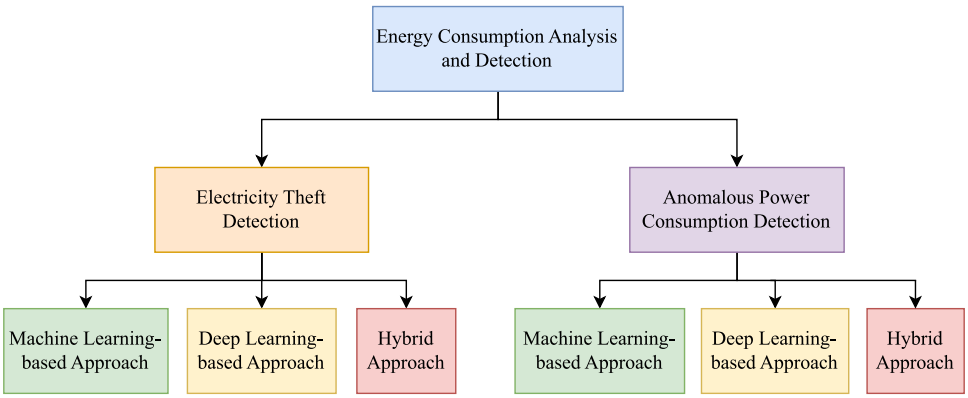



**Fig. 2** Taxonomy of the articles selected for systematic review

neighborhood area network method was employed for smart meter detection. Metrics such as power consumption time interval, correlation, and variance were analyzed for theft detection.

In [15], a statistical framework was proposed to monitor the power consumption of the appliances by disaggregating the overall power consumption. Hidden Markov models and segmented integer quadratic constraint programming was employed to solve the non-intrusive load monitoring

problem by disaggregating a household power profile into the appliance level.

The authors of [16] proposed an appliance load monitoring system using virtual stochastic sensors. A hidden non-Markovian model was employed for the disaggregation process. To evaluate the accuracy of the disaggregation, reconstruction accuracy was calculated. The proposed approach was applied to various household appliances.

The disaggregation technique for appliance power consumption was proposed in [17]. The authors have used active power and reactive power in conjunction with a factorial hidden Markov model. Each appliance-level model was represented using bivariate hidden Markov models. Experiments were conducted on the dataset under both noisy and denoised conditions. Enhanced disaggregation of power consumption for appliance monitoring is proposed in [18]. The authors used harmonic current vectors to develop appliance load signatures.

The simplicity and interpretability of statistical models have made them popular for identifying anomalies and electricity theft. However, in current power consumption scenarios, they exhibit several drawbacks. These models are less effective at representing the complex, non-linear interactions present in large-scale, dynamic energy systems due to their reliance on predetermined assumptions regarding data distribution and linearity.

Statistical methods are less flexible in adapting to changing consumption patterns compared to data-driven models (ML, DL, and hybrid models), limiting their scalability and resilience for practical applications.

## 2.4 Electricity theft detection

### 2.4.1 Machine learning-based approach

ML models can discover anomalous usage patterns; they are being utilized increasingly to detect power theft. The models such as eXtreme Gradient Boosting (XGBoost), Decision Tree (DT), Support Vector Machines (SVM), etc. are able to differentiate between legitimate and fraudulent usage. ML models provide considerable gains over traditional methods in detecting theft by utilizing supervised or unsupervised learning techniques. Their use in SG improves grid security.

An electricity theft detection in Advance Metering Infrastructure (AMI) with XGBoost proposed [19]. Irish Smart Energy Trails dataset was used to train the XGBoost classifier. Six different attack types were used and their proposed approach produced best results on all attack types. In [20], authors have introduced a clustering and local outlier factor-based model for electricity theft detection in AMI. The load patterns were analyzed using k-means clustering and applied local outlier factors to determine the anomaly degrees. A sensitivity analysis was also performed to explore the impact of the number of electricity thieves. Realistic SGCC dataset was used for experiments, that consists of seven different attack types.

An effective way of identifying electricity theft based on similarity measures, a DT combined with K Nearest Neighbors (KNN) and SVM proposed [6]. Regular and power-stealing patterns were identified and more simulated patterns for power-stealing were generated. The system produced an accuracy of 93% and omission ratio of 4%. An ensemble model proposed [5] to detect electricity theft. Adaptive boosting, categorical boosting, extreme boosting, light boosting, Random Forest (RF), and extra trees were included. Statistics-based oversampling was performed to balance the dataset and used different preprocessing methods to improve detection accuracy.

A feature engineering-based ML framework proposed [21] to identify electricity theft. The Synthetic Minority Oversampling Technique (SMOTETomek) was employed for handling data imbalance and used supervised ML techniques like XGBoost, Light Gradient-Boosting Machine (LightGBM), Ensemble bagging and boosting, and other conventional models. Feature extraction and a scalable hypothesis algorithm were used to select the best features. To interpret the decision of the model, the tree-SHapley Additive exPlanations (SHAP) algorithm was utilized. SGCC dataset was used and their proposed approach produced an accuracy of 93.38% and F1-Score of 0.937. The authors of [22] proposed a system based on metaheuristic techniques, including the Artificial Bee Colony (ABC) algorithm and the Genetic Algorithm (GA), to identify electricity theft. By performing statistical analysis, they synthesized 11 new features, which were then used for classification with a SVM. Using the SGCC dataset, the system achieved an accuracy of 87% with ABC and 94% with GA.

To reduce the impact of outliers, a self-decision ant colony clustering algorithm was proposed for detecting electricity theft [23]. The suggested methodology used a dynamic weighting strategy that continuously modifies sample weights according to their significance, in contrast to traditional clustering techniques that update cluster centers using mean values. The classification performance was improved by introducing a novel activation function named"Odd"to handle non-linear correlations. Experimental results show that the algorithm outperforms current techniques in detecting electricity theft by utilizing the SGCC and Smart Data Set for Sustainability (SDSS) datasets.

Using both real and fake attack data, the authors of [24] suggested a ML-based architecture for smart home electricity theft detection. Extreme Gradient Boosting (XGB), RF, and Multilayer Perceptron (MLP) are among the supervised and unsupervised learning models used in the study, which

mimics five assault classes. With the greatest AUC values of 98.69% and 98.74% for simulated and real assault detection, respectively, XGB shows that the framework is robust in identifying electricity theft in a variety of household situations.

A ML-based model for detecting electricity theft was proposed for Indonesian residential postpaid users using traditional meters [25]. The suggested approach employed a sequential evaluation framework that combines RF and KNN to analyze historical electricity usage data. With a 90% F1-Score, 89% accuracy, and 98% recall, the model's detection accuracy was greatly increased by multilayer filtering technique. The outcomes showed how well the system produces trustworthy Target Operation (TO) lists for on-site validation.

To solve class imbalance in the detection of electricity theft, authors of [26] proposed a sample-weighted (SW) loss function. The suggested SW loss ensures higher model performance on minority (theft) class data by assigning different weights based on sample relevance, in contrast to standard loss functions that treat all samples equally. The method was used with different neural network topologies and was independent of model. With an average precision gain of 9.78% and an AUC improvement of 2.92%, experimental results using real-world datasets demonstrated that SW loss performs noticeably better than more conventional techniques like binary cross-entropy loss and class-balanced loss.

In [27], authors introduced an optimal sample selection method to handle the problem of data annotation in supervised electricity theft detection. To enhance detection performance while minimizing labeling costs, three new selection procedures were proposed: diversity-focused, class imbalance-aware, and uncertainty-based. The system chooses the most informative data points for annotation, which greatly increases model efficiency. The study maintains excellent detection reliability while minimizing annotating efforts.

The authors of [28] introduced a strong framework for detecting electricity theft that combines stacked learning models and resampling techniques. They used a mix of near-miss undersampling and borderline synthetic minority oversampling to address the problem of class imbalance. Several ML models were combined with DL-based meta-classifiers like Bi-GRU and Bi-LSTM in a stacked ensemble technique. The suggested ensemble framework considerably increased detection accuracy over conventional techniques, according to experimental results. The study emphasized how hybrid resampling and stacking might improve the effectiveness of electricity theft detection.

#### 2.4.2 Deep learning-based approach

DL models are more accurate and reliable than conventional techniques, making them essential for detecting electricity theft. These models, including Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs), excel at identifying sophisticated patterns of fraudulent behavior by learning relevant information from training data. Additionally, DL models can process both spatial and temporal data, enabling them to detect even subtle abnormalities in power usage. Their integration with SGs enhances security and monitoring capabilities, ensuring reliable and efficient energy delivery.

Authors of [29] have presented a wide and Deep CNN (DCNN) for detecting electricity theft. Instead of 1-dimensional electricity consumption patterns, 2-dimensional data patterns were used by transforming the input data. This transformation was achieved by selecting weekly patterns consisting of seven days instead of using daily patterns. The SGCC dataset was used for the experiments. The system achieved an AUC score of 0.792 and MAP@100 of 0.955. A hybrid convolutional layer and a Long Short-Term Memory (LSTM) was proposed to identify electricity theft [30]. CNN was embeded with LSTM to extract features. The SGCC dataset was utilized for experiments and performance in terms of the accuracy of 96.6% and F1-Score of 0.966 was obtained. In [31], authors have proposed a Deep Neural Network (DNN) for detecting electricity theft. Time and frequency domain features were combined into the DNN. SGCC dataset was used for the experiments and the system achieved an accuracy of 91.8% and AUC score of 0.97. A Graph CNN (GCNN) was introduced for electricity theft detection by employing Euclidean distance measure for graph implementation [32]. Time and latent features were used to detect electricity theft. The authors utilized the SGCC dataset to classify power consumption as either normal or theft, achieving a classification performance with a MAP@100 score of 0.981.

Authors of [33] proposed an enhanced neural network model to identify electricity theft in the SG using SGCC dataset. A SMOTE with an edited nearest neighbor was used to handle data imbalance. AlexNet and GoogLeNet was used for dimensionality reduction and performed the classification using Adaptive Boosting AdaBoost. In [34], authors have proposed an electricity theft detection model based on a stacked sparse autoencoder. Theft behavior was identified based on higher reconstruction error. A DCNN was employed to identify electricity theft for the security of the SG [35].

To improve detection accuracy, the authors of [36] suggested a multi-scale approach for detecting electricity theft that used feature engineering. Electricity usage data is used

to extract features, which were then ranked by XGBoost. The data is subsequently analyzed using a Multi-Scale Convolutional Neural Network (MCNN) using attention mechanisms on various frequency and temporal scales. By efficiently allocating weights to different channels, the attention mechanism minimizes noise and highlights features connected to theft. According to experimental results, the suggested model outperforms conventional techniques, especially when it comes to accuracy and F1-Score.

The study by the authors of [37] presented a brand-new method for detecting electrical theft that combines DL with dual-time feature fusion. They used Temporal Convolutional Networks (TCN) in conjunction with a DCNN and a LSTM (LSTM-TCN) to extract aspects of electricity use at both the daily and weekly levels. A fully linked layer is then utilized to fuse and classify the extracted features. The model's exceptional accuracy was demonstrated by evaluation on the SGCC dataset, where it achieved 94.7% with high precision, recall, F1-Score, and AUC. The model performs better than current detection techniques, and the dual-time feature fusion successfully captures theft trends.

Using a One-Dimensional CNN (1DCNN) tuned with Attention Mechanisms (AM) and Bi-LSTM, the authors of [38] introduced a multi-classification technique for detecting solar electricity theft dubbed 1DCNN-AMBL. The model used an objective function that minimizes the False Positive Rate (FPR) to handle issues caused by uneven data. The proposed approach produced much lower FPR of 0.61% and AUC of 98.46% when compared to other state-of-the-art classifiers on the Ausgrid Power Company dataset. The technique is useful for detecting different kinds of solar electricity theft and minimizing financial losses.

The authors of [39] examined how Energy Theft Detection (ETD) are susceptible to hostile attacks, with a focus on CNNs. To produce adversarial samples that trick ETD models, it presents the Adversarial Data Generation Method (ADGM) and uses the Fast Gradient Sign Method (FGSM). Experiments showed that adversarial attacks drastically lower accuracy when using the CNN-Adaboost model for detection. The results highlight the necessity of creating secure and reliable ETD models in order to lessen antagonistic threats.

To detect electricity theft in three-phase power systems, a detection technique dubbed Catch22-Conv-Transformer was used [40]. It extracts statistical features from multi-dimensional power status data using the Catch22 feature set. Convolutional Neural Networks (CNNs) and a Transformer architecture are used to process these features in order to identify local and global patterns suggestive of theft. Tested on the SGCC dataset, the suggested strategy achieves higher accuracy than traditional methods in detecting theft activities such evasion, tampering, and data modification.

In [41], Dynamic Generative Residual GCNN (DGR-GCNN) proposed as a unique way to identify electricity theft. The suggested model dynamically changes adjacency matrices in real time to reflect changing consumption patterns, in contrast to standard models that have trouble capturing complicated linkages in power consumption data. Even in situations with little theft data, the technique successfully detects electricity theft by incorporating MixHop graph convolution for deeper insights into temporal connections. To improve classification accuracy and handle class imbalance difficulties, the SMOTE is also used. In terms of detection performance, experiments on the SGCC dataset show that the model outperforms traditional models.

A new method for detecting power theft in integrated energy systems that takes into consideration connections between multiple energy loads-such as heating, cooling, and electricity presented in [42]. A Chebyshev Graph Convolutional Network (ChebGCN) is used in the suggested technique to detect malevolent users and capture intricate relationships. Adjacency matrices are created to show correlations when multi-energy loads are represented as graphs, and feature matrices are used to record consumption information. When compared to conventional electricity-only models, experimental results demonstrate that the inclusion of multi-energy load data greatly improves detection accuracy. In both low and high fraud rate scenarios, the ChebGCN continuously performs better than other machine learning models at detecting theft.

A deep one-class classification system was presented [43], which was improved by adding periodicity information from data on power use. The model detects deviations that might point to theft and develops reliable representations of typical electricity use. Compared to traditional one-class approaches, the method increases detection sensitivity by focusing on periodic patterns. The results of the experiments demonstrate that the periodicity increase greatly improves performance in detecting unusual consumption patterns. All things considered, the research shows encouraging outcomes when using deep learning methods to the detection of electrical theft.

To improve the detection of energy theft in areas where distribution shifts occur, the authors of [44] presented a sample adaptive transfer learning technique. The suggested approach takes behavioral differences in electricity consumption into account when adapting labeled data from one city to another. Through the reduction of distributional disparities between datasets, the transfer learning framework enhances detection performance.

#### 2.4.3 Hybrid approach

Hybrid models have shown significant potential in detecting power theft by combining ML and DL approaches. These models leverage the strengths of both ML and DL to enhance efficiency. By handling diverse data formats and extracting intricate patterns, hybrid models are particularly effective at identifying subtle trends in power theft. The integration of ML and DL enables better feature engineering, improved predictive performance, and superior generalization to new data. Consequently, hybrid models play a crucial role in safeguarding SGs and ensuring efficient energy management.

Authors of [45] proposed a hybrid approach by combining AdaBoost, SVM, and General Regression Neural Network (GRNN) to identify electricity theft and the time interval during which the stealing took place. SVM was used to change the weights of the training process in order to attain a particular threshold. Then, identified the theft and estimated their power consumption and duration using the GRNN. To detect electricity theft in a SG, a hybrid model was proposed by combining AlexNet, AdaBoost, and ABC [46]. SGCC dataset was used for the experiments. AlexNet was used to reduce the dimensionality of the dataset. AdaBoost was used to perform the classification. The hyperparameters of these two models were tuned using the ABC algorithm and achieved the accuracy of 88% and F1-Score of 0.78.

In [47], authors have proposed a hybrid model by stacking ML and DL models to analyze electricity theft. SVM, RF, and gradient-boosting decision trees were used as first-order classifiers. Then, the output of these classifiers was used as an input of internal layers DL classifier named TCN to perform final classification. The first-order classifiers were trained by changing the various hyperparameters. The SGCC dataset was utilized for the experiments and obtained an AUC score of 0.985. An intelligent framework to detect electricity theft was proposed [48]. The dynamic time warping-based imputation method was used to fill the missing values and the near-miss undersampling technique to handle data imbalance. The XGBoost and RF were used to perform the final classification. In [49], authors have proposed a big data analytics-based method for electricity theft detection in smart cities. The SGCC and PRECON datasets were used for the experiments. Tomek Link Borderline SMOTE with SVM was utilized for imbalance handling and TCN with Enhanced MLP for classification.

The authors of [50] presented ADASYN-SGWO-PRF, a hybrid approach to power theft detection that combines Adaptive synthetic sampling (ADASYN) with a pipeline RF (PRF) classifier for fraud detection and a Sample Subset Grey Wolf Optimizer (SGWO) for oversampling. This technique improves classification performance by intelligently undersampling and oversampling to correct class imbalance. By removing superfluous features, feature selection with Binomial Cuckoo Search (BCS) further optimizes the model. Additionally, the study uses statistical analysis such as the Friedman and Holm post-hoc tests to validate its efficacy.

To detect electricity theft in SG while protecting privacy, the authors of [51] presented a heterogeneous federated learning system (HeteroFL). To solve class imbalance, the model used a data partitioning and aggregation scheme that gives each class a separate weight, guaranteeing that each class contributes equally. HeteroFL protects the privacy of user data by securing computations on encrypted parameters using the CKKS homomorphic encryption algorithm. Both temporal and spatial interdependence in patterns of electricity usage are captured by a CNN-LSTM model. Experiments on a real-world smart grid dataset show that the framework outperforms traditional methods in identifying electricity theft due to its high accuracy and robustness.

To improve the identification of electricity theft, the integration of gas and electricity usage data was done [52]. In order to integrate gas consumption features into ML models, it suggests two model-agnostic frameworks: the multichannel network framework and the twin network framework. The study showed that adding gas data greatly increases detection accuracy by examining the relationship between patterns of gas and power use. The suggested frameworks perform better than conventional electricity-only models in detecting fraudulent activity, according to experimental results from several datasets.

The issues of incomplete data and class imbalance was addressed by introducing a double-layer stacking optimization framework for electricity theft detection [53]. The technique efficiently fills in missing data and produces balanced synthetic theft samples by using a two-stage TimeGAN for data reconstruction and augmentation. To provide thorough representation during model training, distinct stealing patterns are identified using the K-shape clustering approach. To increase detection accuracy, the stacking architecture also combines several classifiers using a heuristic optimization technique. The usefulness of the framework is demonstrated by experimental results on large-scale power consumption datasets, which show notable improvements in detection rates when compared to baseline models.

The authors of [54] suggested using AI to find hidden connections in order to detect electricity theft. They utilized satellite imagery and AI techniques to detect buildings and match them with registered consumer units. Discrepancies between registered units and detected buildings are resolved using a mathematical assignment method that includes weight calculation strategies.

With little labeled data, a two-stage detection strategy was proposed for detecting electricity theft in new areas [55]. A Deep Subdomain Adaptation Network (DSAN) was used in the initial stage to move knowledge from an existing region to a new one. By choosing representative samples for fine-tuning, the second stage uses Core-Set Active Learning (CSAL) to improve the model. The suggested approach achieves higher detection performance and quick adaption in new regions, according to experimental data.

Tables 1, 2, 3, and 4 summarizes various approaches for detecting theft of electricity.

#### 2.4.4 Comprehensive comparison of electricity theft detection models

The research on electricity theft detection includes various ML, DL, and Hybrid approaches. In ML-based methods, techniques such as Ant Colony Clustering [23] and RF combined with KNN [25] have been used to enhance theft detection through dynamic clustering and sequential evaluation. Advanced loss functions, like Sample-Weighted Loss [26], have been proposed to address class imbalance. Other works [27, 28] focus on optimal sample selection and stacked learning with hybrid resampling to improve classification performance.

DL-based studies incorporate architectures like Multi-Scale CNN [36] and LSTM-TCN with DCNN [37] to extract temporal and spatial features. Some models [38] integrate attention mechanisms and Bi-LSTM for multi-class classification. Others [40, 41] leverage Transformer-based and Graph CNN-based techniques for enhanced feature representation and class balancing.

Hybrid methods blend ML and DL with additional techniques. Federated learning-based models [51] ensure privacy while detecting theft in distributed networks. Time-GAN-based data reconstruction [53] improves prediction reliability, while multimodal approaches [52] integrate gas consumption data to refine detection accuracy. Some studies [54] also incorporate satellite imagery and AI for theft prospecting.

Overall, the advancements emphasize improving accuracy, handling class imbalances, and integrating diverse data sources for robust electricity theft detection.

### 2.5 Anomalous power consumption detection

#### 2.5.1 Machine learning-based approach for anomalous power consumption detection

A technique for dynamic anomaly identification based on data mining was proposed [56]. K-medoid clustering was employed to group the consumption patterns and KNN to

**Table 1** Electricity theft detection

| Reference & Year | Journal (J)/ Conference (C) | Category | Model Used | Key Feature |
|---|---|---|---|---|
| Yan and Wen [19] & 2021 | J | ML | XGBoost | Performed binary classification. Prepared six type of malicious samples. |
| Peng et al. [20] & 2021 | J | ML | K-means and Local Outlier Factor | Introduced seven types of attack patterns. Performed sensitivity analysis. |
| Kong et al. [6] & 2021 | J | ML | KNN and SVM | Identified theft using similarity measure. Generated simulated stealing patterns. |
| Gunturi and Sarkar [5] & 2021 | J | ML | Different boosting models and RF | Performed statistic-based oversampling. Prepared predictive model by ensembling multiple ML models. |
| Hussain et al. [21] & 2021 | J | ML | XGBoost, LightBGM, and some bagging trees | Feature extraction and a scalable hypothesis algorithm was used to select the best features. The tree-SHAP algorithm was used to interpret decisions made by the algorithm. |
| Shehzad et al. [22] & 2022 | J | ML | ABC, Genetic Algorithm, and SVM | Performed statistical analysis and synthesized 11 new features. Proposed system that reduces overfitting, and storage. |
| Zheng et al. [29] & 2017 | J | DL | DCNN | 2D data pattern were used for classification. Performed study on impact of parameters. |
| Gao et al. [30] & 2022 | J | DL | CNN and LSTM | Borderline-SMOTE was used to handle data imbalance issues. Performed k-fold cross-validation to validate performance. |
| Lepolesa et al. [31] & 2022 | J | DL | DNN | Combined time and frequency domain features into the DNN model. PCA was used to reduce the dimension and Bayesian optimization to optimize the parameters. |

match the consumption data. Semi-supervised model was created by combining supervised techniques with clustering. Since it do not identify the cause of the abnormality, it is inappropriate for enhancing building operations. The Particle Swarm Optimization (PSO) was employed for parameter optimization. Point and collective anomalies were identified. A model based on wavelet transform and robust regression was proposed in [57] to detect the anomalies in the load profiles. Wavelet transform was employed at the first stage of detection and robust regression was used as second stage of detection. Errors in the forecasted values were used to decide the anomalies.

Authors of [58] compared 12 statistical and ML algorithms for anomalous power consumption detection. Three datasets and 144 configurations were employed for comparison. The impact of the length of the time series data was also evaluated by using sliding window to detect anomaly. Isolation Forest, one-class SVM, and Local Outlier Factor were used for anomaly detection. In [59], authors have proposed an entropy weight and isolation forest for anomalous power detection. First, the users with similar patterns were grouped using k-means clustering, then reduced the feature set by using Principal Component Analysis (PCA) and entropy weight then weighted isolation forests were employed to identify anomalies.

Fault detection and anomaly identification methods are proposed for the electric power grid [60]. One-class SVM and PCA were used to analyze the impact of feature selection methods. The PSO was used for optimizing the hyperparameters. Authors of [61] have used federated learning in distributed anomaly detection in SGs. The local smart meter data was used for training. In [62], authors have used a Rain Flow-based Mean Nearest Neighbor Distance Anomaly Factor algorithm to detect abnormal power consumption. Based on the threshold, point and pattern anomalies were identified.

In [63], authors used different models like Prophet, LightGBM, and Vector Autoregressive model to perform forecasting and anomaly detection. Household appliance dataset was used to perform anomaly detection and forecasting. The Improved KNN (IKNN) was employed to categorize unexpected power utilization [64]. Two approaches have been used: supervised anomaly detection with micromoments and unsupervised outlier detection. A simulated dataset was used, labeled with five different categories, to classify the power consumption of appliances as either normal or anomalous. Authors of [65] outlined a method based on data analytics that can locate and characterize anomalies in daily energy patterns. Multiple clustering was used like “Follow the Leader"clustering strategy. Load patterns were identified and grouped using Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and K-means clustering. A method for identifying anomalies in the daily energy consumption patterns of central air conditioning systems has been developed [66]. Hierarchical clustering was employed and the information entropy characteristic value was used to identify patterns in energy consumption.

Authors of [67] have suggested a model to identify abnormal energy use in data from smart meters using contextual analysis. Four anomaly detection heuristics were selected where each heuristic used different methods to find the abnormalities. KNN and PCA were employed for detection purpose. Authors in [68] have looked into the difficulties presented by annotated datasets and performance evaluation measures. A statistical and linear regression strategy were used for automatic labeling.

To identify irregular electricity use, an unsupervised anomaly detection model that combines an Isolation Forest with an Improved Canopy-K-Means algorithm was proposed [69]. Weighted density improvements are added to the Canopy-K-Means clustering algorithm to maximize cluster formation, while PCA feature fusion aids in dimensionality reduction. An Isolation Forest model is used in the last anomaly detection step to improve classification accuracy. When evaluated on data from smart meters, the suggested method outperforms conventional clustering techniques in terms of precision and recall.

To find outliers in Mexico’s household electricity usage, anomaly detection methods was proposed in [70] by using AI-based approaches. The three main detection models that are used are Median Absolute Deviation (MAD), Local Outlier Factor (LOF), and Isolation Forest. Electricity usage is predicted by a predictive ML-based approach over a 30-day period, and discrepancies were detected by using anomaly detection. The findings show that 75% of anomalies are successfully captured by Isolation Forest, and model outputs are interpreted using Shapley Additive Explanations (SHAP) analysis. The study improves the identification and justification of unusual electricity use, which aids in smart city energy management.

To discover possible reasons and detect unusual variations in building energy usage, a computational intelligence method was proposed [71]. The study established consumption patterns for different buildings in Ecuador, Spain, France, and Canada using a statistical confidence interval analysis. A machine learning-based classification algorithm is used to identify anomalies, improving detection accuracy over the conventional Seasonality Analysis of Electricity Consumption (SAECC) method. The suggested approach is more successful in detecting contextual anomalies in power usage. The results encourage zero-energy buildings to implement data-driven energy efficiency measures.

An anomaly detection technique based on unsupervised learning was presented in [72] to detect unusual power

**Table 2** Electricity theft detection contd

| Reference & Year | Journal (J)/ Conference (C) | Category | Model Used | Key Feature |
|---|---|---|---|---|
| Liao et al. [32] & 2022 | J | DL | GCNN | Euclidean distance measure was used for graph implementation. |
| Aldegheishem et al. [33] & 2021 | J | DL | AlexNet and GoogLeNet | SMOTE with nearest neighbors was used to handle imbalance. A conditional Wasserstein generative adversarial network with a gradient penalty was used to capture the features. |
| Huang and Xu [34] & 2021 | J | DL | Sparse Autoencoder | An autoencoder was utilized to learn the effective features. Identified theft behavior based on higher reconstruction error. Particle Swarm Optimization was used to optimize the hyperparameters. |
| Haq et al. [35] & 2023 | C | DL | DCNN | A five-layered CNN was utilized to classify the power consumers. |
| Wu et al. [45] & 2018 | C | Hybrid | AdaBoost, SVM, and GRNN | SVM was used to change the weights. GRNN was used for classification. |
| Ullah et al. [46] & 2022 | J | Hybrid | AlexNet, AdaBoost, and ABC | Near miss technique was used for undersampling, AlexNet was utilized to reduce the dimensionality, and AdaBoost was used to perform the classification. Hyperparameters were tuned using ABC. |
| Khan et al. [47] & 2021 | J | Hybrid | RF, SVM, and TCN | SMOTE and Tomek-l. link methods were used to handle the data imbalance. ML models were used as first order and DL as second order classifiers. |
| Kulkarni et al. [48] & 2021 | C | Hybrid | Autoencoder, RF, and SVM | Dynamic time warping-based Imputation method was used to fill the missing values and near miss technique for handling imbalance issue. |
| Arif et al. [49] & 2022 | J | Hybrid | SVM, TCN, and MLP | SMOTETomek was used for handling imbalance. Average ensemble strategy was used to avoid high variance in the DL models. |

usage patterns in residential users. To improve detection accuracy, the model breaks down electricity consumption into 2-hour time chunks. The primary anomaly detection technique, Local Outlier Factor (LOF), focuses on density-based outlier identification. To increase the resilience of the model, data preprocessing techniques like standardization and feature extraction are used. The results show that unsupervised ML can be used to identify unusual trends in electricity consumption.

The authors of [73] proposed unsupervised clustering approaches to solve the issue of anomaly detection in the electricity usage of public street lighting. Data on energy use is gathered by using the ENEA PELL smart city platform, and aberrant behaviors are detected using the k-means, DBSCAN, and OPTICS clustering algorithms. The study offers a scalable method for spotting irregularities in extensive public lighting networks, which aids in smart city energy management.

Using a Gaussian Mixture Model (GMM) and the Multi-Cluster Feature Selection (MCFS), an unsupervised method for detecting abnormal power usage was proposed [74]. To describe patterns of electricity consumption, the technique extracts twelve essential elements from power load sequences, such as morphological markers and fluctuation indices. In order to improve clustering performance, the MCFS algorithm is used to minimize dimensionality while maintaining multi-cluster topologies. On the basis of probabilistic modeling, GMM was used to distinguish between normal and atypical power consumers. Using actual data from Anhui Province, China, the experimental results show that the suggested approach works better in terms of recall rate, precision, and F1-Score than both K-means clustering and conventional PCA-based feature selection.

A ML framework for anomaly detection and energy consumption prediction in smart homes was proposed in [75]. Using three outlier detection techniques (Local Outlier Factor, Connectivity-based Outlier Factor, and Cluster-Based Local Outlier Factor) for anomaly detection and a variety of ML models (Linear Regression, RF, Decision Tree, KNN, and Gradient Boosting) for energy consumption prediction are the two stages of the framework. It shows that XGBoost and KNN perform better than other models in predicting energy use by evaluating the framework on three publicly accessible datasets. The findings suggest that combining predictive modeling and anomaly detection improves energy efficiency in smart homes.

In [76], XGBoost Location, Scale, and Shape (XGBoost-LSS) and Ensemble Batch Prediction Intervals (EnbPI), two probabilistic techniques for identifying systemic abnormalities in building energy usage, were compared. The purpose of the study is to assess how well these models forecast energy usage and use an alarm matrix to spot reoccurring

**Table 3** Electricity theft detection contd

| Reference & Year | Journal (J)/ Conference (C) | Category | Model Used | Key Feature |
|---|---|---|---|---|
| Yang [23] & 2024 | J | ML | Ant Colony Clustering | Self-decision clustering with dynamic weighting and dropout mechanism for theft detection. |
| Abraham et al. [24] & 2024 | J | ML | XGB, RF, MLP | Detection using real and synthetic attack data with improved accuracy in smart homes. |
| Taruna et al. [25] & 2025 | J | ML | RF + KNN | Sequential evaluation framework using RF and KNN for theft detection in traditional meter users. |
| Liao et al. [26] & 2025 | J | ML | SW Loss | Sample-weighted loss function for improved class imbalance handling in theft detection. |
| Liao et al. [27] & 2024 | J | ML | Optimal sample selection | Reduces annotation efforts using uncertainty-based and diversity-aware strategies. |
| Ullah et al. [28] & 2025 | J | ML | Stacked learning with hybrid resampling | Combines oversampling, undersampling, and ensemble learning for theft detection. |
| Zhang [36] & 2024 | J | DL | MCNN | Feature engineering with XGBoost for theft detection using multi-scale convolution and attention mechanism. |
| Huang [37] & 2024 | J | DL | LSTM-TCN + DCNN | Dual-time feature fusion using LSTM-TCN and DCNN for improved theft detection accuracy. |
| Chen [38] & 2024 | J | DL | IDCNN-AMBL | Multi-classification with attention mechanism and Bi-LSTM to reduce false positive rate. |
| Nirmal [39] & 2025 | J | DL | CNN-Adaboost | Adversarial attacks using ADGM and FGSM to evaluate ETD model robustness. |
| Bai [40] & 2024 | J | DL | Catch22-Conv-Transformer | Multi-dimensional feature extraction using CNN and Transformer for three-phase theft detection. |
| Zhuang et al. [41] & 2024 | J | DL | DGR-GCNN | Dynamic graph convolutional network using MixHop and SMOTE for theft detection. |

abnormalities. While both models produce precise point and probabilistic predictions, XGBoostLSS outperforms EnbPI in terms of accuracy, while EnbPI produces greater recall rates but more false alarms. According to a robustness examination, XGBoostLSS stays stable but EnbPI performs better when trained with noisy data. The study emphasizes how crucial probabilistic forecasting is for identifying collective anomalies and enhancing building energy efficiency.

### 2.5.2 Deep learning-based approach for anomalous power consumption detection

The GoogLeNet and ResNet, and their combination GoogLeResNet3, has been utilized [77] to detect the customer's anomalous power consumption behavior. Inception and residual modules have been included to handle the gradient divergence problem. In [78], authors have proposed a combined spatio-temporal-based anomaly detection method. Temporal information was extracted by using TCN and spatial features using Graph Neural Network (GNN). An anomaly detection method was proposed for a power system [79]. Bidirectional LSTM (Bi-LSTM) was combined with Graph Convolutional Network (GCN) for the detection process.

Authors of [80] employed CNN and Bi-LSTM for energy prediction task and based on prediction error the anomalies were identified. A smart meter system that identifies anomalous power consumption was proposed in [81]. Bi-LSTM auto-encoder was employed for anomaly detection. Based on the reconstruction error, the anomalies were identified. In [82], authors have proposed DNN for five class anomalous power consumption classification. Authors of [3] have suggested a method for anomaly detection and power consumption prediction using an auto-encoder.

A multi-agent-based anomaly identification method proposed [83]. Ensemble method was employed to tag the data, and then used LSTM and an auto-encoder to find the anomalies. Authors of [84] have used Symbolic Aggregate Approximation based on DNN to predict future regular and irregular behaviors and the pattern of electricity usage in a building. A labeled parent repository was prepared for anomaly identification, and anomalies are identified by comparing pattern frequency with the discord threshold.

A secure Industrial Internet of Things (IIoT) infrastructure that incorporates anomaly detection and AI-driven load forecasting for intelligent energy management was presented in [85]. The suggested model used an improved Isolation Forest for anomaly detection and a Temporal Convolutional Network (TCN)-GRU-Attention architecture for short-term active and reactive load forecasting. With TLS/SSL protocols and encrypted credentials, the system is built for real-time deployment on edge and cloud servers,

**Table 4** Electricity theft detection contd

| Reference & Year | Journal (J)/ Conference (C) | Category | Model Used | Key Feature |
|---|---|---|---|---|
| Liao et al. [42] & 2025 | J | DL | ChebGCN | Multi-energy load analysis using graph convolution for theft detection in integrated energy systems. |
| Wu and Wang [43] & 2025 | J | DL | PE-DOCC | Periodicity-enhanced deep one-class classification model. |
| Liao et al. [44] & 2024 | J | DL | Sample Adaptive Transfer Learning | Transfers labeled data while adapting to distribution shifts in electricity theft detection. |
| Tripathi [50] & 2024 | J | Hybrid | ADASYN-SGWO-PRF | Hybrid adaptive sampling and pipeline random forest for improved electricity theft detection. |
| Wen [51] & 2025 | J | Hybrid | HeteroFL | Privacy-preserving federated learning using CNN-LSTM with homomorphic encryption. |
| Liao [52] & 2024 | J | Hybrid | Multichannel and Twin Network | Gas consumption data integration for enhanced electricity theft detection. |
| Ge [53] & 2025 | J | Hybrid | Double-Layer Stacking + TimeGAN | Data reconstruction using TimeGAN with stacking optimization for theft detection. |
| Kaminski et al. [54] & 2025 | J | Hybrid | AI-based building detection with assignment algorithm | Uses satellite imagery and AI for theft prospecting. |
| Wang et al. [55] & 2024 | J | Hybrid | Two-stage screening with hybrid modeling | Combines domain adaptation and active learning for generalizable detection. |

guaranteeing scalability and security. For real-time anomaly detection and energy forecasting, the suggested IIoT architecture offers a scalable, safe, and effective solution.

To identify unusual electricity usage in household appliances, authors of [86] presented a proactive anomaly detection framework that makes use of Variational Autoencoders (VAE). A dynamic thresholding approach that adjusts to changes in power usage patterns improves the suggested CNN-LSTM-VAE model. To improve model performance, the methodology uses Lowess smoothing and additive decomposition, as well as a feature engineering process that fine-tunes appliance operation classifications. By avoiding malfunctions and inefficiencies, testing on 26 front-loading washing machines shows a 30% reduction in energy use. The study offers a brand-new unsupervised learning strategy for raising the energy efficiency of household appliances.

To identify irregularities in smart grid energy usage, authors in [87] presented a trace-based graph deep learning algorithm. The model used LSTM and a Graph Neural Network (GNN) to combine the temporal and spatial characteristics of smart meter data. In order to encapsulate consumption patterns and connections inside the power distribution network, it creates a Trace Quality Graph (TQG). Using an unsupervised encoder-decoder architecture, the model extracts spatial features using variational graph autoencoders (VGAE) and temporal features using LSTM autoencoders. Hyperparameters that balance temporal and spatial loss values are used to calculate anomaly scores. With an F1-score of 94.60% and an AUC of 98.90%, experimental findings on a real-world dataset showed superiority over conventional anomaly detection algorithms.

To detect unusual periods of power use in non-intrusive load monitoring (NILM), an anomaly-detection-based learning architecture was proposed in [88]. For anomaly detection, the method combines the Multivariate State Estimation Technique (MSET), Local Outlier Factor, and LSTM. The standard for comparison is the Hidden Markov Model (HMM). It enhances real-time evaluation of power usage irregularities by efficiently identifying cyber-intrusions and possible privacy risks in smart grids. The findings show that LSTM works better than alternative techniques in precisely identifying irregularities in data on electricity usage, allowing smart power systems to be more secure and effective.

An enhanced Deep Variational Autoencoder (DVAE) was proposed to model patterns of electricity usage and identify anomalies [89]. This method improves anomaly detection in commercial and industrial electricity usage by combining sliding window adaption with auto-time series updating. Southeast Chinese and U.S. Department of Energy datasets are used to train the model.

### 2.5.3 Hybrid approach

The advantages of both ML and DL techniques are combined in hybrid models to offer a comprehensive solution for anomalous power usage detection. The load forecaster was designed to utilize the benefits of both ML model like k-means clustering and DL models like DNN and LSTM. The LSTM-Gauss-NBayes approach, which integrated with LSTM, Bayesian, and Gaussian methods to detect the anomaly in Industrial IoT [90]. Naive Bayes (NB) was employed to see if it could detect the anomaly using an error testing set.

In [91], authors have offered anomaly identification on home appliances by using variational autoencoder to detect anomaly. The performance was evaluated with one-class SVM to assess the efficacy of their suggested work. An autoencoder was used for extracting features of anomalous power consumption in households [92]. One-class SVM was trained on the extracted features, and performed anomaly detection.

Authors of [93] have proposed real-time anomaly detection in electricity load using hyper-dimensional computing. ML models like SVM and KNN, and DL models like DNN and LSTM were utilized for experiments. In [94], authors have employed an image classification model based on transfer learning. Pre-trained models such as SqueezeNet, AlexNet, and GoogleNet were used for feature extraction and SVM was employed as a classifier. In [95], authors have proposed a hybrid data mining approach by using RNN and Quantile Regression (QR) to discover anomalies in residential buildings' energy use.

Authors of [96] have suggested K-means clustering and LSTM-based anomaly detection in a SG. Utilizing the cluster centroid, the LSTM model was trained to predict consumption for the following day. The anticipated consumption was considered typical if the predicted value falls within the range ($-2\sigma$ to $+2\sigma$). In [97], authors have provided a technique for anomalous power consumption identification by employing Feed-forward DNN and one-class SVM. Using a Raspberry Pi prototype, typical activity patterns were collected.

The authors of [98] proposed, a two-step end-to-end methodology, for identifying unusual patterns in power usage. The first stage clustered similar consumption patterns using Grey Wolf Optimization (GWO), while the second stage refined anomaly identification using a Generative Adversarial Network (RGAN) based on a Relational Stacked Denoising Autoencoder (RSDAE). The model tackles classification issues brought on by noisy inputs and efficiently extracts features from high-dimensional data. According to experimental assessments, GC-RGAN outperforms traditional models like RF and SVM in detecting electricity theft. This work demonstrates the advantages of using semi-supervised, adversarial learning, and clustering approaches to enhance the detection of electrical theft.

In [99], a mix of ML algorithms was used to offer a dynamic anomaly detection method for building energy usage. The authors used Classification and Regression Trees (CART) for data identification, Entropy-Weighted k-means (EWKM) for clustering, and a hybrid strategy that combines local outlier factor and Isolation Forest (IF) for anomaly detection. Collective anomalies are detected via Dynamic Time Warping (DTW). The approach outperforms existing models in terms of accuracy and is validated using actual building energy data. The study shows how well hybrid models work for energy management and anomaly detection.

A unified graph-based data analytics system for knowledge transmission, anomaly detection, and clustering in intelligent energy networks was presented in [100]. The suggested approach uses a network embedding to represent multivariate time-series data, with nodes standing for patterns of energy usage and edges for inter-dependencies. While graph-based anomaly detection techniques uncover possible defects and abnormal consumption patterns, spectral clustering is used to identify customer groupings. By integrating expert knowledge into the graph structure, the framework also incorporates knowledge propagation to increase detection accuracy. Real-world district heating and cooling networks are used to validate the method, which shows enhanced anomaly detection accuracy and interpretability.

The AF-GS-RandomForest anomaly detection approach was proposed in [101] to find outliers in time-series power consumption data. Modules for detection and prediction make up the model. Autoformer, a DL technique, was used in the prediction module to foresee nonlinear and unstable sequences. The detection module looks for notable residual deviations using a grid-search optimized RF method. When tested on datasets from actual office buildings, the method outperforms traditional methods in terms of precision, recall, and F1-Scores. The findings show that combining ML-based anomaly detection with DL-based time series forecasting improves the accuracy and resilience of identifying unusual energy usage.

An ensemble framework for identifying unusual energy use in industrial buildings was presented in [102]. To find subsequence anomalies, the framework combines the Local Outlier Factor, Deep Isolation Forest (DIF), and Anomaly Transformer (AT). To increase the dependability of the data, missing values are imputed using a transformer-based model.

Tables 5, 6, 7, 8, and 9 show the summary of various anomalous power consumption detection approaches.

**Table 5** Anomalous power consumption detection

| Reference & Year | Journal (J)/Conference (C) | Category | Model Used | Key Feature |
|---|---|---|---|---|
| Lei et al. [56] & 2023 | J | ML | KNN and Medoid clustering | Employed PSO to tune parameters. |
| Karkhaneh and Ozgoli [57] & 2022 | J | ML | Wavelet transform and regression | Errors in the forecasted values were used to decide the anomalies. |
| Zangrando et al. [58] & 2022 | J | ML | Isolation Forest, One-class SVM, and Local Outlier Factor | Compared 12 statistical and ML models. Three datasets and 144 configurations were used for comparison. |
| Jianyuan et al. [59] & 2022 | J | ML | K-means clustering and Isolation Forest | Reduced the feature set by using PCA and weighted Isolation Forest was used for identifying anomalies. |
| Elmasry and Wadi [60] & 2022 | J | ML | One-class SVM | Fault and anomaly detection was carried out with dimensionality reduction by using PCA. |
| Jithish et al. [61] & 2023 | J | ML | Federated Learning with ML | Performed distributed anomaly detection without violating the data security of the consumer. |
| Yin et al. [62] & 2022 | J | ML | Rain Flow-based Mean Nearest Neighbor Distance Anomaly Factor | Performed parameter turning for the unsupervised models to improve accuracy. |
| Malki et al. [63] & 2022 | J | ML | Prophet, Light GBM, and Vector Autoregressive model | Anomaly on IoT environment identified based on forecasting error. |
| Himeur et al. [64] & 2021 | J | ML | IKNN | Performed supervised anomaly detection with micro-moments. |
| Piscitelli et al. [65] & 2021 | J | ML | DBSCAN and K-means clustering | Multiple clustering with"Follow the Leader"strategy was used to ascertain anomalous power consumption patterns. |
| Zhou et al. [66] & 2021 | J | ML | Hierarchical clustering | Information entropy characteristic value was used to identify patterns in energy consumption. |

Tables show different models used for detection purposes with the key features of the respective approaches.

#### 2.5.4 Comprehensive comparison of anomalous power consumption detection models

The reviewed studies explore various ML, DL, and hybrid approaches for detecting anomalies in power consumption. ML-based methods primarily employ clustering (e.g., K-means, DBSCAN, OPTICS), outlier detection (e.g., LOF, Isolation Forest), and ensemble models (e.g., XGBoost, KNN). Feature selection and dimensionality reduction techniques, such as MCFS-GMM, have also been incorporated to improve model performance.

DL-based approaches frequently integrate LSTM, CNN, and autoencoders for time-series anomaly detection, leveraging real-time forecasting and attention mechanisms. Models like TCN-GRU-Attention and CNN-LSTM-VAE introduce novel architectures for proactive anomaly identification in smart grids and home appliances.

Hybrid methods combine ML and DL models to enhance anomaly detection accuracy. These include graph-based clustering, autoencoders, and ensemble learning techniques such as Autoformer-RF and LOF-DIF-Anomaly Transformer. Some studies, like GC-RGAN and EWKM-CART-IF-LOF, focus on semi-supervised and rule-based techniques for better generalization.

Overall, the research trend highlights a shift towards combining multiple learning paradigms to enhance anomaly detection robustness in power consumption analysis.

### 2.6 Privacy-preserving methods

Privacy preservation in SG is essential because of the sensitive nature of the data acquired, particularly energy consumption patterns from individual residences or companies. Safeguarding the privacy rights of persons, preventing unauthorized access, and mitigating security threats such as cyber attacks and data breaches all depend on protecting this data. Furthermore, privacy-preserving procedures assist in addressing issues with behavioral profiling, adhering to privacy requirements legally, and upholding public acceptability and confidence. SG operators may assure the ethical and responsible use of data, cultivate strong customer connections, and encourage the widespread use of energy-efficient technology by prioritizing privacy protection.

Authors of [104] presented a brand-new privacy-preserving federated learning architecture for identifying energy theft. A secure protocol enabling detection stations was created to transmit encrypted training parameters. Homomorphic encryption was used to compute the aggregated parameters. TCN was utilized to identify energy theft. To ensure privacy in electricity theft detection, authors [105] have proposed a blockchain-enabled energy theft detection system. Security was maintained so malicious users could not alter the energy consumption data, and no unauthorized user should be aware of the consumer's energy usage.

In [106], authors have offered a thorough overview of ML-based model developments in detecting false data attacks in SGs. The data sources and SG architecture, the types of fake data assaults, and presented data security specifications were also summarized. In [107], authors have proposed an efficient electricity theft detection model with privacy preservation. To protect consumers' privacy, an efficient scheme was proposed by encrypting smart meter readings. Authors of [108] presented a model for identifying energy theft with privacy preservation by utilizing CNN and Paillier algorithm. An effective and privacy-preserving power theft detection technique proposed in [109] for the AMI network. Their strategy protected users' privacy through secret sharing while enabling system operators to detect power thefts, monitor loads, and quickly calculate electricity bills using disguised fine-grained meter data.

### 2.7 Federated learning for anomalous power and electricity theft detection

Federated learning has been a viable paradigm in recent years for resolving issues with scalability and data privacy in anomaly detection systems. Federated learning, in contrast to conventional centralized methods, allows cooperative training of machine learning models across dispersed devices without necessitating the sharing of raw data. Since sensitive user data from smart meters and IoT devices poses privacy threats, this decentralized architecture is especially pertinent in the areas of power consumption and electricity theft detection.

In [110], authors have proposed a stacked ensemble GRU with optimization to classify cyberattacks and to detect electricity theft using federated learning. Different statistical approaches were used to identify optimal features, and LSTM to fill missing values and achieved an accuracy of 95% and showed remarkable interpolation. It also identified nine types of cyberattacks. GRU provided a lightweight structure and reduced the computation costs.

Authors of [111] proposed a lightweight model for anomaly detection in a time series dataset using a federated learning approach. Datasets related to industrial control systems and other time series data sets, including univariate data sets on power demand, were used for processing. They compared their proposed federated learning approach with the centralized approach and demonstrated that the federated learning approach outperformed the centralized one. Even though the behavior related to normal and abnormal

**Table 6** Anomalous power consumption detection contd

| Reference & Year | Journal (J)/Conference (C) | Category | Model Used | Key Feature |
|---|---|---|---|---|
| Sial et al. [67] & 2021 | J | ML | KNN and PCA | Contextual analysis was used to identify anomalies by using eighteen groups and four heuristics. |
| Gaur et al. [68] & 2019 | J | ML | Linear Regression | Automatic labeling of anomalies was performed and different performance assessment criteria were discussed. |
| Yin and Zhong [77] & 2023 | J | DL | GoogLeNet and ResNet combined as GoogLeResNet3 | Included Inception and residual modules to handle the gradient divergence problem. |
| Kong et al. [78] & 2023 | J | DL | TCN and GNN | Proposed spatio-temporal-based anomaly detection method. Temporal information were learned using TCN and spatial features using GNN. |
| Lee et al. [79] & 2023 | J | DL | Bi-LSTM and GCN | Combined two DL models to improve the performance and experiments were conducted by using 900 household dataset. |
| Pan et al. [80] & 2022 | J | DL | CNN and LSTM | High-dimensional power consumption data was used for the experiments. Forecast the energy and analyzed the anomaly based on prediction error. |
| Lee et al. [81] & 2022 | C | DL | Bi-LSTM and Autoencoder | Anomalies were identified based on reconstruction error. |
| Himeur et al. [82] & 2020 | J | DL | DNN | Experiments were conducted by using three labeled datasets and analyzed the impact of various hyperparameters. |
| Chahla et al. [3] & 2020 | J | DL | Autoencoder | Anomalies were identified based on reconstruction error. |
| Weng et al. [83] & 2018 | J | DL | LSTM and Autoencoder | Introduced the multi-agent-based anomaly identification. |
| Lin et al. [84] & 2022 | J | DL | DNN | Symbolic Aggregate Approximation was used to identify patterns and labeled parent repository was prepared for anomaly identification. |

activity changes with time, the federated approach could be able to adapt to these changes as frequent updates used to take place in the federated learning approach.

A hybrid DL model proposed for electricity theft detection using federated learning [112]. The convolutional GRU model was used to capture spatial and temporal features. The authors compared their proposed approach with models such as CNN, LSTM, and GRU in a federated learning environment.

In [113], authors proposed a privacy-protected framework for detecting electricity theft based on federated learning. Their main aim was to ensure the security and detection of electricity theft in a distributed environment. The CNN was used for theft detection and performed parameter analysis as well. Also comparison analysis was performed with RNN, LSTM, and GRU in the federated learning environment.

In [114], the electricity theft detection approach was proposed using federated learning. The authors have ensured the privacy concerns about user data through federated learning. Traditional ML models such as RF, KNN, and bagging classifiers were used to classify power consumption as normal or theft activity. The authors also proposed a unique voting classifier that combines the classification results of various ML classifiers and produces the final outcome based on a majority voting procedure.

Authors of [115] proposed a federated learning-based anomaly detection system using an autoencoder. Through their proposed approach, they have addressed problems like delay in response time and the problem of data leakage. Their proposed system provided high accuracy in detecting anomalies in distributed systems. Dynamic threshold selection was employed alongside the peak over threshold algorithm to automatically determine the threshold value for anomaly detection. The effectiveness of the proposed approach was demonstrated through comparisons with state-of-the-art methods.

In [116], the authors proposed an electricity theft detection model using a federated learning approach. Smart energy IoT data was utilized for the experiments. A CNN model operated within the federated learning environment to detect electricity theft. Data imbalance was addressed using SMOTE, with a primary focus on increasing the true positive rate of theft detection.

In [117], a federated learning-based electricity theft detection model was proposed. The authors proposed a functional encryption cryptosystem to address limitations such as privacy preservation, aggregation, high overhead, and low accuracy. A hybrid approach combining CNN and GRU models was employed. Detection rate and false alarm were considered as performance evaluation metrics, alongside accuracy.

A deep electricity theft detection framework based on federated learning to detect zero-day cyberattacks was proposed in [118]. Deep autoencoder architecture was employed for electricity theft detection. The performance was compared against Isolation Forest and OCSVM. An unsupervised approach was adopted for theft detection. Six different types of cyber attacks were considered, and the performance of the proposed federated learning approach was compared with that of a centralized approach.

Privacy preservation is a key feature of the federated learning approach. User data does not need to be sent to a centralized system, as it is processed on edge devices through distributed processing. By leveraging federated learning, utility providers can collaboratively develop reliable detection models while safeguarding user privacy and complying with data protection regulations. The approach's ability to handle heterogeneous data generated by diverse energy consumption patterns across regions further enhances its applicability in real-world scenarios.

## 3 Dataset used in electricity theft and anomalous power consumption detection

Datasets play a crucial role in research on electricity theft and anomalous power consumption detection, serving as the foundation for developing, training, and testing predictive models. Electricity theft datasets primarily focus on capturing patterns of unauthorized energy usage and often include labeled data distinguishing theft and non-theft scenarios. In contrast, anomalous power consumption datasets aim to identify irregular energy usage patterns caused by equipment failures, behavioral anomalies, or external factors. These datasets may consist of time-series energy readings, voltage, current, and metadata about users or appliances, typically derived from real-world measurements, simulated scenarios, or hybrid approaches combining both. Despite their importance, this field faces several challenges, including the scarcity of large, real-world datasets, concerns about data privacy, and the lack of established standards.

The datasets commonly used in electricity theft detection are listed in Table 10, along with details such as the number of consumers, the frequency of power consumption readings, and the duration for which the readings are taken.

The identification of abnormal power usage has been a key research topic, supported by numerous datasets specifically designed for this purpose. These datasets contain energy usage patterns across various appliances and locations, covering a wide range of scenarios, from domestic to commercial settings. The data is carefully curated, often with labeled information to distinguish between typical and unusual consumption patterns. Using these datasets,

**Table 7** Anomalous power consumption detection contd

| Reference & Year | Journal (J)/Conference (C) | Category | Model Used | Key Feature |
|---|---|---|---|---|
| Wang et al. [103] & 2023 | J | Hybrid | K-means, DNN, and LSTM | Designed load forecaster by utilizing the benefits of both ML and DL models. Created a look-back optimizer and a training data generator for forecasting. |
| Wu et al. [90] & 2019 | J | Hybrid | LSTM, Bayesian, and Gaussian | Tested NB an ML-based model to see if it could detect the anomaly using an error testing set. |
| Castangia et al. [91] & 2022 | J | Hybrid | Variational Autoencoder and One-class SVM | Offered anomaly identification for home appliances. By computing the training error, the anomaly threshold was decided. |
| Tsai et al. [92] & 2022 | J | Hybrid | Autoencoder and One-class SVM | Autoencoder was used for extracting features of anomalous power consumption and One-class SVM was trained on extracted features. |
| Wang et al. [93] & 2022 | J | Hybrid | SVM, KNN, DNN, and LSTM | Proposed a real-time anomaly detection in electricity load using hyper-dimensional computing. designed retraining functions to improve classification accuracy. |
| Copiaco et al. [94] & 2023 | J | Hybrid | SqueezeNet, AlexNet, GoogleNet, ans SVM | Used transfer learning to extract features and SVM for classification |
| Xu et al. [95] & 2020 | J | Hybrid | RNN and Quantile Regression | Proposed a hybrid data mining approach to discover anomalies in residential buildings' energy use. |
| Fenza et al. [96] & 2019 | J | Hybrid | K-means and LSTM | Trained LSTM based on cluster centroid and anomalies were identified by setting a threshold as 2 time sigma. |
| Nimmy et al. [97] & 2023 | J | Hybrid | Feed Forward DNN and One-class SVM | Used Raspberry Pi prototype to collect activity patterns and produced evidence of a Distributed Denial of Service (DDoS) attack on energy usage. |

advanced detection models can be developed and benchmarked. The datasets vary in complexity and include temporal, geographical, and contextual information. The datasets commonly used in anomalous power consumption detection are listed in Table 11, along with details such as the number of consumers, the frequency of power consumption readings, the duration for which readings are collected, and the number of appliances involved.

## 4 Nature of anomalies

Based on the nature of the anomalies, they are classified into three categories namely, point anomalies, contextual anomalies, and collective anomalies.

- Point Anomalies: These are single data points that differ considerably from the remaining points in the dataset. An abrupt and significant rise in electrical energy consumption at an odd period, like a spike in electricity use late at night, would be regarded as a point anomaly in power usage. Finding problems like equipment failures or unauthorized use requires the detection of such anomalies. Machine learning techniques, such as Random Forests, have been used in recent research to forecast building energy use and identify outliers by analyzing prediction residuals [142].
- Contextual Anomalies: These anomalies arise when a data point is regarded as normal in one context but abnormal in another. For instance, heavy power usage in the winter might be typical because of heating requirements, but it would be unusual to use the same amount of power in the summer. Models that take seasonal fluctuations and temporal trends into account are necessary for detecting contextual anomalies. DL models, like Autoencoder, have been used in recent studies to forecast time-series data and spot abnormalities by analyzing prediction residuals [142].
- Collective Anomalies: These involve a collection of data points that, even though they may not be anomalous individually when taken as a whole, deviate from the typical pattern. A slow leak in the system or a malfunctioning device could be the cause of a string of days-long, progressive rises in power use. Analyzing data point sequences to find patterns that depart from accepted standards is a common step in detecting collective abnormalities. To discover anomalies in time-series data, recent research has used DL approaches, including LSTM networks, to model and forecast patterns of energy consumption [143].

**Table 8** Anomalous power consumption detection

| Reference & Year | Journal (J)/ Conference (C) | Category | Model Used | Key Feature |
|---|---|---|---|---|
| Wang and Li [69] & 2024 | J | ML | Improved Canopy-KMeans + Isolation Forest | Combines clustering and Isolation Forest for unsupervised anomaly detection. |
| Solís-Villarreal et al. [70] & 2024 | J | ML | IF-LOF-MAD | A predictive AI-based approach using outlier detection for energy consumption anomalies. |
| Luna-Romero et al. [71] & 2024 | J | ML | CI-SAECC | A computational intelligence-based approach improving anomaly detection accuracy in electrical consumption. |
| Ho et al. [72] & 2024 | J | ML | LOF | A time-segment anomaly detection model using density-based outlier detection. |
| Ali et al. [73] & 2024 | J | ML | K-means-DBSCAN-OPTICS | A clustering-based approach for anomaly detection in public street lighting energy data. |
| Liu et al. [74] & 2024 | J | ML | MCFS-GMM | A feature selection and clustering approach for unsupervised abnormal power consumption detection. |
| Ambat and Sahoo [75] & 2024 | J | ML | LOF-COF-CBLOF + XGBoost-KNN | A combined anomaly detection and prediction framework for smart home energy consumption. |
| Stjelja et al. [76] & 2024 | J | ML | EnbPI-XGBoostLSS | A probabilistic anomaly detection approach using prediction intervals and alarm matrix. |
| Joha et al. [85] & 2024 | J | DL | TCN-GRU-Attention | Integrates real-time load forecasting and anomaly detection in a secure IIoT framework. |
| Papaioannou et al. [86] & 2024 | J | DL | CNN-LSTM-VAE | Uses dynamic thresholding and VAE for proactive home appliance anomaly detection. |

Since each kind requires specialized detection techniques to precisely identify and handle possible concerns, understanding these classifications is crucial for creating anomaly detection systems in power consumption.

## 5 Applications of electricity theft and anomalous power consumption detection

In order to promote more sustainable and economical building operations, detection of anomalous power consumption and electricity theft provide valuable applications for building management, energy efficiency, and operational optimization. Use of anomalous energy consumption monitoring in buildings is expanding beyond energy efficiency and is now being applied in various innovative scenarios. Specifically, they might be employed to identify (i) occupancy in the building, (ii) elderly monitoring at home, (iii) Unusual power consumption behavior, (iv) malfunctioning appliances, and (v) non-technical losses. Furthermore, without needing additional systems, an anomaly detection system within a structure may be utilized for numerous purposes, such as detecting non-technical losses or occupancy. As a result, this may successfully lower the costs associated with implementing hardware and simplify established systems.

**Identifying occupancy in the building** Buildings can optimize energy use, save waste, and improve occupant comfort by correctly identifying occupancy levels. It will ultimately result in considerable energy savings and increased operational efficiency. Furthermore, occupancy data makes predictive analytics and adaptive control techniques possible, allowing for more proactive resource management in buildings and improved tenant requirements and preferences alignment. Actual occupancy detection needs the installation of various sensors and cameras, which costs more and requires dedicated hardware and software installments [144].

Authors of [145] detected the occupancy using ML classifiers based on electricity usage data. The data collection includes sensor data from smart power outlets, passive infrared (PIR) sensors, and a household's electricity use. In [146], authors have used SVM and KNN to identify contextual occupancy for an office. Contextual nature of electricity information was utilized to identify an extra state when employees leave the office for small breaks. In [147], authors proposed an occupancy detection technique by employing SVM for a smart home using appliance power usage data. For occupancy identification, their method combines aggregated and appliance-specific

**Table 9** Anomalous power consumption detection

| Reference & Year | Journal (J)/ Conference (C) | Category | Model Used | Key Feature |
|---|---|---|---|---|
| Ida Evangeline et al. [87] & 2024 | J | DL | GNN-LSTM-VGAE | A trace-based deep learning model integrating spatial and temporal attributes for anomaly detection. |
| Chen et al. [88] & 2024 | J | DL | LSTM-LOF-MSET | A deep learning-based anomaly detection approach for non-intrusive load monitoring in smart grids. |
| Lin et al. [89] & 2024 | J | DL | DVAE | A variational autoencoder-based model with a sliding window for anomaly detection in energy consumption. |
| Aslam et al. [98] & 2024 | J | Hybrid | GC-RGAN | Uses GWO clustering and RGAN for semi-supervised anomaly detection. |
| Lei et al. [99] & 2024 | J | Hybrid | EWKM-CART-IF-LOF | A hybrid method combining clustering, decision trees, and anomaly detection for building energy consumption. |
| Grassi et al. [100] & 2024 | J | Hybrid | Graph-based Clustering | A graph embedding approach integrating clustering, anomaly detection, and knowledge propagation. |
| Mao et al. [101] & 2024 | J | Hybrid | Autoformer + RF | A DL-based forecasting model combined with a ML classifier to detect anomalies in power consumption. |
| Song et al. [102] & 2024 | J | Hybrid | LOF + DIF + Anomaly Transformer | An ensemble framework integrating multiple unsupervised learning techniques to detect subsequence anomalies. |

power consumption data. In [148], authors have proposed a technique for occupancy sensor anomaly classification based on a lighting system. One-class RF, regular RF, and K-means clustering were employed for detection of occupancy anomaly.

**Elderly monitoring at home** Authors of [149] proposed an elderly habit monitoring model based on electricity consumption. Their method consisted of three distinct acquisition portions: an Android mobile application for remote data access and a distant server for data processing and storage. An approach to monitor the behavior of older people at home using appliance power consumption data by integrating the Personal Data Capturing System of the City4 Age Platform was proposed [150]. In [151], authors presented a household appliance usage-based elderly monitoring system. The approach relied on a discreet smart meter infrastructure that continuously monitors the total power used in the home and a few smart plugs for device-specific monitoring.

**Detecting unusual power consumption behavior** In [82], authors have identified unusual behavior of the consumer by analyzing the power consumption readings and utilizing DNN to classify the unusual power consumption. Authors of [152] proposed rule-based system to identify abnormal power consumption using data mining. A complete suite of building sensor knowledge was used to design the data mining system. Authors of [153] put forth an innovative strategy with ML that takes power usage of a smart home device and identifies anomalous power consumption. DDoS assaults were then launched to record changes in the power profile.

**Identifying malfunctioning appliances** The issue of detecting defective appliances is addressed by the approach called Rimor ([154]). Using contextual information and previous energy data, it estimated a home's energy use and flagged an anomaly when the actual energy consumption differed considerably from the expected consumption. Authors of [155] evaluated NILM algorithms to monitor appliance-level anomalies. The efficacy of NILM was assessed for anomaly identification and suggested a supervised anomaly detection method. In order to identify the locations of probable energy frauds and malfunctioning meters, authors od [156] proposed a novel framework by utilizing linear programming for anomaly detection that assesses users' energy utilization behavior. Two new metrics were introduced: the loss factor and the error term.

**Identifying non-technical losses** In [157], authors have proposed an electricity fraud detection model for the SG. SVM was used to identify energy fraud. Best feature list

**Table 10** Some of the datasets used in electricity theft detection

| Dataset | # Consumers | Duration | Sampling Granularity |
|---|---|---|---|
| UMass Smart [119] | 400+ households | 2 years | 1 minute |
| SGCC [120] | 42372 | 2 years | Daily |
| Irish Smart Energy Trial [121] | 5000+ households and industrial consumers | 535 days | 30 minutes |
| Individual household electric power consumption [122] | One household | 4 years | 1 minute |
| Pecan Street [123] | 1000+ | - | Real Time |
| AER [124] | 25 households | 2 years | 30 minutes |
| Electricity Load Diagrams [125] | 370 | 4 years | 1 hour |
| Low Carbon London Smart Meter Trial [126] | 5567 households | 2 years | 30 minutes |

**Table 11** Some of the datasets used in anomalous power consumption detection

| Dataset | # Households | Duration | Sampling Frequency (Hz) | # Devices used |
|---|---|---|---|---|
| REDD [127] | 6 | 119 days | 1/3 | 24 |
| HES [128] | 26 | 1 year | 1/600 | 23 |
| BLUED [129] | 1 | 1 week | 12 k | - |
| iAWE [130] | 1 | 73 days | 1 | 10 |
| ECO [131] | 6 | 8 months | 1 | - |
| AMPds1 [132] | 1 | 1 year | 1/60 | 21 |
| UK-DALE [133] | 4 | 2 years | 1/6 | 5 |
| DRED [134] | 1 | 6 months | 1/60 | 13 |
| GREEND [135] | 8 | 1 year | 1 | 9 |
| HUE [136] | 1 | 3 years | 1/60 | - |
| BLOND [137] | 1 | 213 days | 1/6 | 16 |
| REFIT [138] | 20 | 213 days | 1 | 9 |
| AMPds2 [139] | 1 | 2 years | 1/60 | 21 |
| PSD [140] | 10 | 1 week | 1/60 | - |
| SMARTEN-ERGY [141] | 2 | 1 year | 1/120 | 25 |

was selected from the consumer's energy consumption profile for the prediction. Authors in [158], employed gradient boosting to detect electricity fraud. XGBoost, Logistic Regression, and Isolation Forest were employed for detection purposes. In [159], authors have proposed a rule-based classification method for electricity theft detection. It was proposed to identify anomalies in consumer energy consumption profiles. Hierarchical clustering and a decision tree classifier were employed for the detection. Authors of [160] suggested a Deep RNN for energy theft identification. The suggested model implements a Gated Recurrent Unit (GRU)-RNN using the time series nature of the customer's power usage, enhancing the detection performance. In [161], authors proposed a framework for identifying energy theft attacks in SGs. A theft detector was developed based on the Kullback-Leibler divergence. Tables 12 and 13 summarizes the applications of electricity theft and anomalous power consumption detection. It covers five different applications that benefit from electricity theft and anomalous power consumption detection.

## 6 Gaps in the existing works

Despite using data-driven models, such as ML, DL, and hybrid models, to identify anomalous power usage and electricity theft, which has several significant gaps which need to be addressed.

1. **Validation of Real-time data:**
   - Many models are created and evaluated using small or artificial real-world datasets, which could not accurately reflect the wide range of circumstances prevalent in various geographic and socioeconomic situations. This restricts how broadly the results may be applied.
   - There are no long period studies that track these models' performance over time. These investigations are essential to comprehending the models' long-term dependability and flexibility.
2. **Advanced Techniques:**
   - The field of power theft detection and anomaly detection needs to give more attention to emerging technologies like blockchain, federated learning, and edge computing. These technologies improve model resilience, processing efficiency, and data security.
   - The integration of the technical fields, such as the Internet of Things (IoT) and ML and DL, still needs to be better understood. These multidisciplinary methods could provide novel approaches to critical problems.
3. **Explainability and Interpretability of the Models:**
   - Research on creating interpretable and explicable AI models for these tasks needs to be more extensive.

**Table 12** Application of electricity theft detection and anomalous power consumption detection

| Reference & Year | Journal (J)/Conference (C) | Model Used | Application | Key Feature |
|---|---|---|---|---|
| Kleiminger et al. [145] & 2013 | C | ML | Occupancy Identification | PIR sensors were used to collect data and ML models to identify occupancy |
| Akbar et al. [146] & 2015 | C | SVM and KNN | Occupancy Identification | ML models were used to identify energy patterns. Performed occupancy identification in office. |
| Gao et al. [147] & 2018 | C | ML | Occupancy Identification | Aggregated and appliance power usage data was utilized to identify occupancy and SVM for classification. |
| Violatto et al. [148] & 2019 | J | One-class RF, RF, and K-means | Occupancy Identification | Performed binary classification to identify occupancy and clustering was used to identify similar energy patterns. |
| Visconti et al. [149] & 2019 | C | Sensor-based Approach | Elderly Monitoring | Mobile application and server were used for data collection. Monitored elderly behavior using power consumption data. |
| Patrono et al. [150] & 2017 | C | Rule-based algorithm | Elderly Monitoring | The appliance power usage was monitored to detect elderly behavior |
| Patrono et al. [151] & 2018 | C | Sensor-based Approach | Elderly Monitoring | The strategy was converted into practice and verified under the H2020 City4 Age Project. |
| Himeur et al. [82] & 2020 | J | DL | Detecting unusual power consumption behavior | DNN was used for anomalous power consumption patterns. |
| Peña et al. [152] & 2016 | J | ML | Detecting unusual power consumption behavior | A complete suite of building sensor knowledge to design the data mining system. |
| Dilraj et al. [153] & 2019 | C | ML | Detecting unusual power consumption behavior | Created the illusion of a smart house using brute-force tactics and smart cameras. Launched DDoS assaults to record changes in the power profile. |

Improving these models' transparency is essential to winning over stakeholders and making practical deployment easier.
- There has yet to be much progress in creating user-friendly interfaces that let non-expert users engage with these models and comprehend outcomes. Enhancing usability and accessibility can encourage broader adoption.

4. **Deployment and Scaling:**

- Existing models sometimes need help scaling to enormous datasets characteristic of real-world SG scenarios. Further research on scalable algorithms and structures is required to close this gap.
- The research must fully address practical deployment techniques for these models, including real-time processing, computing resource considerations, and integration with existing infrastructure.

5. **Regulatory Compliance and Ethics:**

- Further investigation is required to guarantee data privacy and adherence to regulatory requirements while utilizing these models. Balancing the security of sensitive customer data and the advantages of sophisticated analytics is a crucial difficulty.
- One area that needs further research is ensuring that models are equitable and impartial for all demographic groups. The research should include methods for identifying and reducing biases in data and model predictions.

## 7 Challenges and limitations

Several significant challenges and limitations exist in identifying abnormal power use and detecting electricity theft. These problems are a result of the particular needs of the applications as well as the intrinsic complexity of the data. Among the main challenges are the lack of annotated datasets, which makes it challenging to train and validate models, and dataset imbalances, which cause biased model performance. The definition of an anomaly might also differ, making it more challenging to create models relevant to all situations. Additionally, the requirement for high granularity in data gathering and processing poses a barrier to identifying abnormalities at the appliance level. Lastly, defining anomalies as normal or abnormal ignores the range of potential anomaly kinds and their associated consequences, oversimplifying the issue.

**Absence of annotated datasets** One major obstacle in conducting research in detecting abnormal power use and electricity theft is the lack of annotated benchmark datasets. Annotated datasets are essential for training and assessing performance of the AI-based models. They provide precisely labeled instances of both normal and aberrant energy usage patterns. These datasets are required for researchers to evaluate the generalization capacities of their detection models and test their performance over a range of contexts. Building strong and robust detection systems is further complicated by the manual annotation process's labor-intensive, time-consuming, and human-error-prone nature. Utility companies, academic institutions, and regulatory agencies must work together to gather, annotate, and disseminate meaningful datasets that accurately reflect the complexity and diversity of actual energy usage patterns to meet this challenge.

**Imbalance in the dataset** The imbalance dataset that used to train the model seriously hinders detection ability of the model. Imbalanced datasets can result in biased model training and subpar performance in identifying minority classes if one class (normal energy consumption, for example) is noticeably more common than others (anomalous consumption, for example). This imbalance in the context of energy management might result from abnormal occurrences occurring less frequently than typical consumption patterns. Because of this, AI-based models trained on imbalanced datasets could not be as sensitive to anomalies, which might result in cases of electricity theft or irregular power use being unnoticed. Careful preparation methods, such as oversampling, undersampling, or artificial data synthesis, are needed to overcome this obstacle.

**Definition of anomaly** The definition of anomalies in detecting anomalous power usage and electricity theft has several difficulties affecting detection systems' efficacy. First, it might be challenging to precisely define anomalies to capture all possible abnormalities in energy usage patterns since anomalies can differ significantly in their traits and expressions. The variety of anomalies necessitates that detection algorithms be flexible and able to adjust to many kinds of abnormalities, such as abrupt spikes, extended deviations, and mild oscillations. Anomalies can also happen occasionally or intermittently, leading to imbalanced datasets with a higher proportion of normal cases than anomalous ones. This can make it difficult to train and evaluate models. Anomaly detection attempts are further complicated by the dynamic nature of energy use trends and the changing strategies used by those who steal electricity.

**Detecting appliance-level anomalies** Finding abnormalities in energy consumption data is made more difficult by the difficulties associated with identifying anomalies at the

**Table 13** Application of electricity theft detection and anomalous power consumption detection contd

| Reference & Year | Journal (J)/ Conference (C) | Model Used | Application | Key Feature |
|---|---|---|---|---|
| Rashid et al. [154] & 2018 | C | ML | Identifying malfunctioning appliances | Introduced Rimor to detect defective appliance. Used simple-to-collect appliance power ratings to identify the aberrant items or appliances. |
| Rashid et al. [155] & 2019 | C | ML | Identifying malfunctioning appliances | Assessed the efficacy of NILM for anomaly identification. |
| Yip et al. [156] & 2018 | J | Linear Programming | Identifying malfunctioning appliances | Proposed framework for anomaly detection that assesses user's energy utilization behavior. Introduced two new metrics the loss factor and the error term. |
| Amara korba et al. [157] & 2019 | C | ML | Identifying non-technical losses | SVM was used to identify energy fraud and selected the best feature list from the consumer's energy consumption profile. |
| Albiero et al. [158] & 2019 | C | ML | Identifying non-technical losses | XGBoost, Logistic Regression, and Isolation Forest were used for detection purposes. Employed gradient boosting and anomaly identification to detect electricity fraud. |
| Jain et al. [159] & 2019 | J | ML | Identifying non-technical losses | Proposed a rule-based classification method for electricity theft detection. Hierarchical clustering and a DT classifier were used for theft detection. |
| Nabil et al. [160] & 2018 | C | DL | Identifying non-technical losses | GRU-RNN for identification. Used random tuning for parameter tuning. |
| Krishna et al. [161] & 2016 | C | Kullback-Leibler Divergence | Identifying non-technical losses | Developed a theft detector based on the Kullback-Leibler Divergence. Assessed the detector using fictitious data injections based on actual smart meter data. |

appliance level. First, it might be challenging to discern between typical appliance functioning and abnormal behavior due to the sheer volume and complexity of data supplied by smart meters and IoT devices. The slight variations in energy usage patterns that appliance-level abnormalities may display make them hard to spot among the noise and unpredictability included in energy consumption statistics. Anomaly detection methods are further complicated by the variety of equipment and their varied energy consumption profiles, as individual appliances may display unique usage patterns and abnormal behavior. Establishing baseline models for anomaly identification is difficult because of the dynamic nature of appliance usage, including occupancy variations, user behavior, and environmental factors. This adds complexity and variability to data on energy consumption. In addition, the absence of labeled datasets with instances of anomalies at the appliance level makes it more difficult to create and assess detection algorithms, which, in turn, reduces the efficacy of anomaly detection systems in spotting and addressing anomalous energy use patterns at the appliance level.

**Classifying anomalies just as normal or anomalous** Due to its simple approach to anomaly characterization, categorizing anomalies as normal or anomalous restricts the identification of abnormal power use. Detection systems may miss the richness and diversity of abnormal behaviors shown in energy consumption data if anomalies are classified into binary types. This oversimplified categorization system misses the subtleties and variances in anomalous patterns, which might result in false positives or false negatives in the detection results. Furthermore, classifying all anomalies as a single category ignores the hierarchical structure of anomalies, which may call for various response mechanisms or detection tactics depending on the kind or level of anomaly. Anomalies, for instance, might range in their degree of severity, length, or effect on the functioning of the energy system, requiring distinct approaches according to the relative risk of each abnormality.

## 7.1 Data quality issues

The accuracy and reliability of anomaly detection models for power consumption analysis are heavily dependent on data quality. Poor data quality can lead to misclassification of anomalies, thereby reducing the effectiveness of detection systems. Common issues, such as missing data, noise, and outliers, significantly impact the detection of abnormal power usage.

Missing data is a common issue in power consumption datasets, often caused by data logging problems, communication breakdowns, or sensor failures. As a result, the

continuity of time-series data may be disrupted, making it challenging for models to identify patterns or trends essential for detecting anomalies. For instance, gaps in a smart meter's data may prevent the identification of a spike in power usage due to a faulty device. To address this, effective imputation techniques, such as linear interpolation or more advanced approaches like deep learning-based generative models, are required. These methods preserve the integrity of temporal patterns while filling in missing values. If missing data is not properly handled, models may exhibit biased learning, leading to false positives or negatives in anomaly detection.

Noise in data refers to random or irrelevant variations in power consumption readings that obscure real usage patterns. Environmental factors, sensor errors, and external interference can all introduce noise. For example, small, sporadic variations in readings caused by electrical interference can mask genuine anomalies, such as a sudden spike in energy usage. If noisy data is not addressed, models may overfit to irrelevant fluctuations, compromising their ability to detect anomalies and generalize effectively. Noise reduction techniques, such as wavelet transforms and smoothing filters, are commonly applied during preprocessing to enhance data quality. Robust algorithms, such as wavelet-based neural networks, are particularly effective in distinguishing between noise and actual anomalies in power data.

Outliers in power consumption data are larger values that significantly differ from the rest of the dataset. While anomalies are a subset of outliers, not all outliers are anomalies; some may result from unusual but legitimate events or sensor errors. For example, an unexpected power surge caused by lightning might be recorded as an anomaly but does not indicate abnormal consumption patterns. Conversely, an outlier representing abnormally high energy use during off-peak hours could suggest equipment failure or electricity theft. Since such outliers can distort model predictions and complicate the detection of true anomalies, it is crucial to identify and manage them. Techniques like isolation forests and robust regression are commonly employed to handle outliers effectively while maintaining the integrity of the dataset.

Building reliable models for abnormal power consumption detection requires addressing data quality issues. Preprocessing techniques such as imputation, noise reduction, and outlier management can significantly enhance the quality of input data. Additionally, leveraging domain knowledge-such as understanding the typical power usage of various appliances-can help differentiate between true anomalies and minor deviations. To improve the robustness of detection systems in real-world applications, future research could focus on integrating automated data-cleaning methods with anomaly detection frameworks.

## 8 Future directions

Future research initiatives are essential to solve the complex and changing issues of electricity theft and abnormal power usage detection. These initiatives include developing real-time detection and response systems, integrating multi-source data fusion approaches, and investigating fine-grained anomaly detection.

- **Fine-grained Anomaly Detection:** One can examine methods for identifying irregularities at a more granular level, such as the level of each appliance or specific periods.
- **Multi-source Data Fusion:** To improve the precision of anomaly detection systems, one can investigate methods for combining data from several sources, such as weather related sensors.
- **Sophisticated ML Techniques:** Researchers can examine sophisticated ML approaches for anomaly identification in energy consumption data. These techniques have to provide more precise and trustworthy results.
- **Real-time Detection and Response:** One can provide systems for real-time anomaly detection that can recognize and react to anomalies as they happen.
- **Privacy-preserving Techniques:** Researchers can examine anomaly detection strategies to integrity and security of private energy usage data while allowing for the efficient identification of anomalies. This might safeguard personal information while facilitating anomaly detection amongst many stakeholders.
- **Evaluation and Benchmarking:** Benchmark datasets and additional standard evaluation metrics can be provided to evaluate the performance of different models.
- **Interdisciplinary Research:** Multidisciplinary cooperation amongst academics in computer science, electrical engineering, sociology, economics, and policy studies can be encouraged to tackle the intricate socio-technical problems of detecting unusual power usage and electricity theft.

## 9 Conclusion

With an emphasis on identifying electricity theft and anomalous power usage, this survey study thoroughly reviews cutting-edge energy consumption analysis and detection. After conducting a thorough literature analysis, we have categorized current methods using data-driven models, such as ML, DL, and hybrid approaches for electricity theft and abnormal power usage detection. Our review demonstrates the wide variety of approaches and algorithms applied in this subject, along with each one's advantages and disadvantages. Brief

summary was provided on the datasets used in electricity theft and anomalous power consumption detection. We have also covered the applications, gaps, challenges, and constraints related to detecting abnormal power usage and electricity theft, highlighting the significance of resolving these concerns to improve energy systems' effectiveness, dependability, and security. In addition, we have identified future research paths that will advance the state-of-the-art in the analysis and detection of anomalous energy consumption. These include real-time detection and response systems, multi-source data fusion, and fine-grained anomaly detection. Through synthesizing insights from previous research and providing recommendations for future investigations, this survey report seeks to support the development of creative solutions to difficulties connected to energy usage in the modern era.

## Appendices

**Table 14** List of Abbreviations used in this article

| List of Abbreviations | |
|---|---|
| Abbreviation | Meaning |
| ABC | Artificial Bee Colony |
| AdaBoost | Adaptive Boosting |
| AMI | Advance Metering Infrastructure |
| Bi-LSTM | Bidirectional-Long Short-Term Memory |
| CNN | Convolutional Neural Network |
| DCNN | Deep Convolutional Neural Network |
| DDoS | Distributed Denial of Service |
| DL | Deep Learning |
| DNN | Deep Neural Network |
| GCN | Graph Convolutional Network |
| GCNN | Graph Convolutional Neural Network |
| GNN | Graph Neural Networks |
| GRNN | General Regression Neural Network |
| GRU | Gated Recurrent Unit |
| IKNN | Improved KNN |
| KNN | K-Nearest Neighbors |
| Light-GBM | Light Gradient-Boosting Machine |
| LSTM | Long Short-Term Memory |
| ML | Machine Learning |
| MLP | Multi-Layer Perceptron |
| NB | Naïve Bayes |
| NILM | Non-Intrusive Load Monitoring |
| PCA | Principal Component Analysis |
| PSO | Particle Swarm Optimization |
| QR | Quantile Regression |
| RF | Random Forest |
| RNN | Recurrent Neural Network |
| SG | Smart Grid |
| SMOTE | Synthetic Minority Oversampling Technique |
| SVM | Support Vector Machine |
| TCN | Temporal Convolutional Network |
| XGBoost | Extreme Gradient Boosting |

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**Dataset Availability** Data will be made available on request.

## Declarations

**Compliance with Ethical Standards** This article does not contain any studies with human or animal subjects.

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